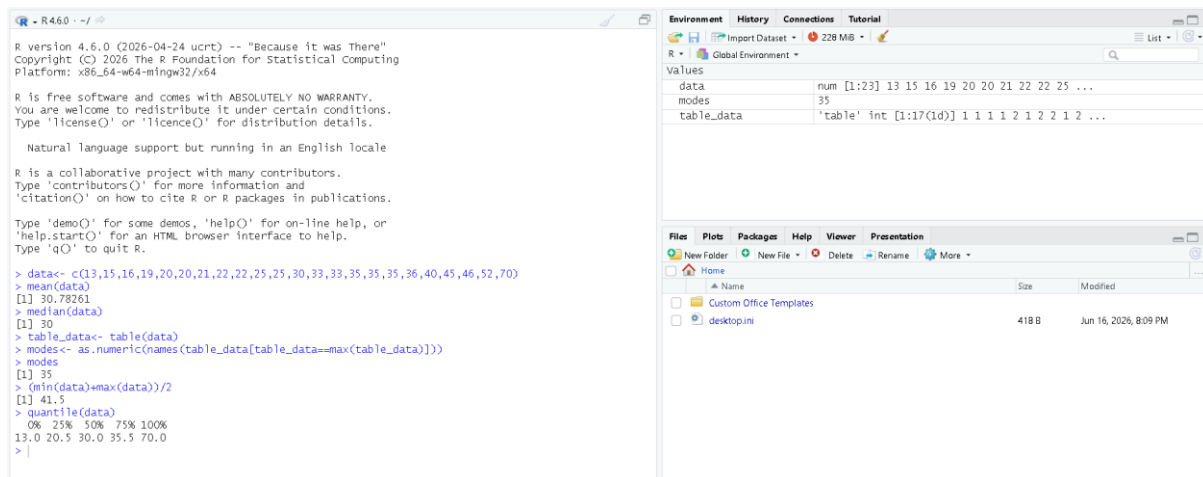


1)



2)

'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

```
> data <- c(200,300,400,600,1000)
> min_value <- min(data)
> mox_value <- max(data)
> minmax<-(data-min_value)/(mox_value-minmax)
```

Error: object 'minmax' not found

```
> minmax <- (data - min_value)/(mox_value - min_value)
> minmax
[1] 0.000 0.125 0.250 0.500 1.000
> zscore <- scale(data)
> zscore
      [,1]
[1,] -0.9486833
[2,] -0.6324555
[3,] -0.3162278
[4,]  0.3162278
[5,]  1.5811388
attr(,"scaled:center")
[1] 500
attr(,"scaled:scale")
[1] 316.2278
> |
```

3)

```
R 4.6.0
> data <- c(12,18,25,27,30,35,42,48,55,60,75,90,120,150,200)
> cut(data,breaks=3)
 [1] (11.8,74.7] (11.8,74.7] (11.8,74.7] (11.8,74.7] (11.8,74.7]
 [6] (11.8,74.7] (11.8,74.7] (11.8,74.7] (11.8,74.7] (11.8,74.7]
[11] (74.7,137] (74.7,137] (74.7,137] (137,200] (137,200]
Levels: (11.8,74.7] (74.7,137] (137,200]
> quantile(data,probs=seq(0,1,length=4))
 0% 33.33333% 66.66667% 100%
12.00000 33.33333 65.00000 200.00000
> kmeans(data,centers=3)
K-means clustering with 3 clusters of sizes 2, 10, 3

Cluster means:
 [1]
 1 175.0
 2 35.2
 3 95.0

Clustering vector:
 [1] 2 2 2 2 2 2 2 2 2 2 3 3 3 1 1

Within cluster sum of squares by cluster:
 [1] 1250.0 2249.6 1050.0
 (between_SS / total_SS = 88.7 %)

Available components:

 [1] "cluster" "centers" "totss" "withinss" "tot.withinss" "betweenss"
 [7] "size" "iter" "ifault"
>
```

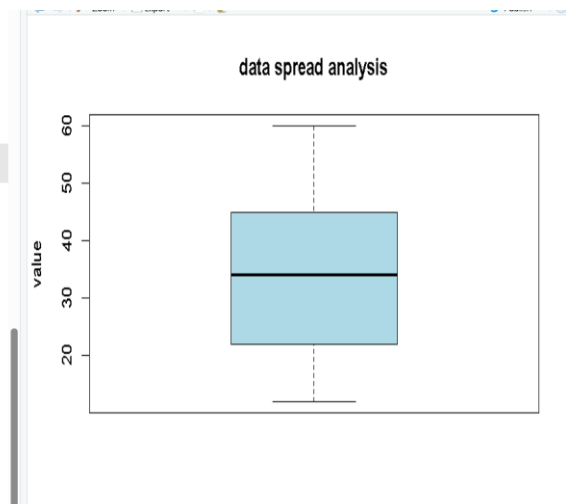
4)

```
>
> data <- c(12,15,22,25,30,38,40,45,52,60)
> quantile(data,probs= c(0.25,0.50,0.75))
 25%   50%   75%
22.75 34.00 43.75
> summary(data)
    Min. 1st Qu.  Median     Mean 3rd Qu.     Max.
 12.00   22.75   34.00   33.90   43.75   60.00
> iqr(data)
```

Error in iqr(data) : could not find function "iqr"

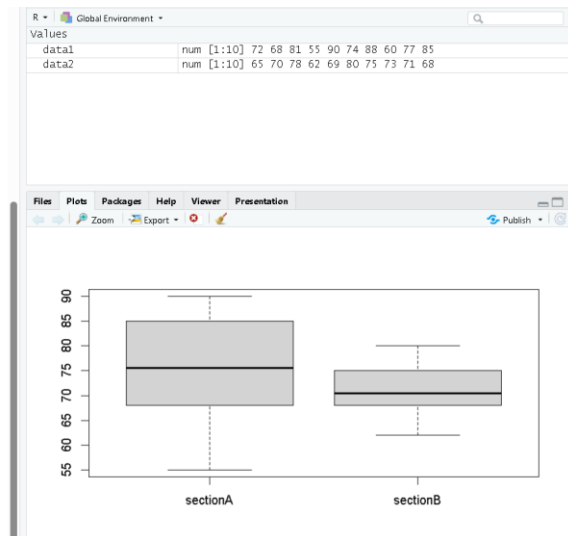
Show Traceback
Rerun with Debug

```
> IQR(data)
[1] 21
> boxplot(data,main="data spread analysis", ylab="value", col="lightblue")
>
>
>
>
>
>
>
>
>
>
>
```



5)

```
> data1 <- c(72,68,81,55,90,74,88,60,77,85)
> data2 <- c(65,70,78,62,69,80,75,73,71,68)
> mean(data1);median(data1);range(data1)
[1] 75
[1] 75.5
[1] 55 90
> mean(data2);median(data2);range(data2)
[1] 71.1
[1] 70.5
[1] 62 80
> boxplot(data1,data2,names=c("sectionA","sectionB"))
>
>
>
>
>
>
>
>
>
>
>
```



6)

```
Copyright (c) 2006 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64

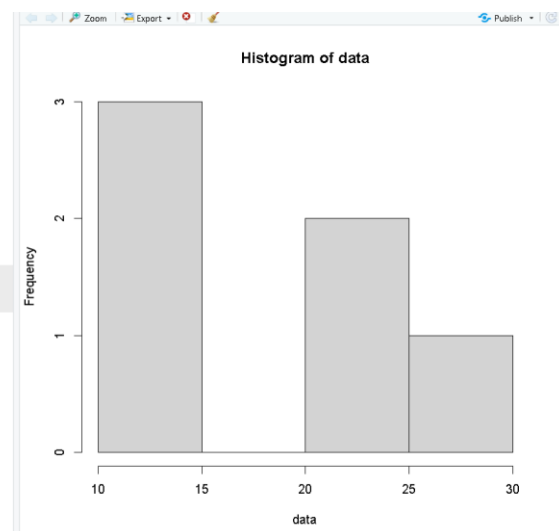
R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

  Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

Error in download.file(url = "https://cdn.posit.co/posit-ai/manifest.json", :
  cannot open URL 'https://cdn.posit.co/posit-ai/manifest.json'
In addition:
  Warning message:
In download.file(url = "https://cdn.posit.co/posit-ai/manifest.json", :
  URL 'https://cdn.posit.co/posit-ai/manifest.json': status was 'could not resolve hostname'
```



7)

```
> num1 = as.integer(readline(prompt="enter a number:"))
enter a number:12
> num2 = as.integer(readline(prompt="enter a number:"))
enter a number:24
> num3 = num1+ num2
> print(num3)
[1] 36
> |
```

8)

```
> num1 = as.integer(readline(prompt="enter a number:"))
enter a number:33
> num2 = as.integer(readline(prompt="enter a number:"))
```

```
Error in readline(prompt = "enter a number:") :
  unused argument (prompt = "enter a number:")
```

```
> num2 = as.integer(readline(prompt="enter a number:"))
enter a number:22
> num3 = num1-num2
> print(num3)
[1] 11
> |
```

9)

```
>
> num1 = as.integer(readline(prompt="enter a number:"))
enter a number:21
> num2 = as.integer(readline(prompt="enter a number:"))
enter a number:12
> num3 = num1*num2
> print(num3)
[1] 252
> |
```

10),

```
> num1 = as.integer(readline(prompt="enter a nummber:"))
enter a nummber:22
> num2 = as.integer(readline(prompt="enter a number:"))
enter a number:11
> num3 = num1/num2
> print(num3)
[1] 2
> |
```